



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°6_ – ADMIN-SYSTEME

COURS: ADMINISTRATION DES SYSTÈMES
INFORMATIQUES

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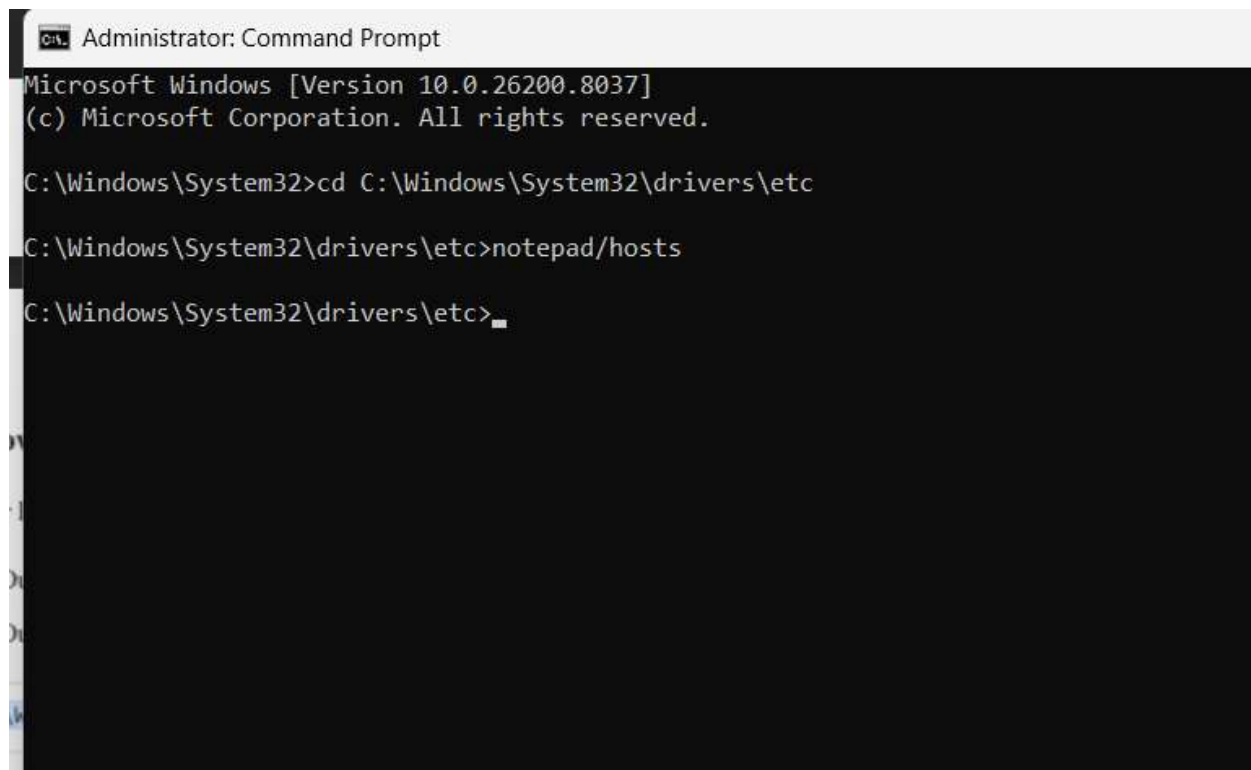
DATE: 12/04/2026

L'objectif de ce TD est de:

1. Déployer une architecture web résiliente
2. Configurer un site dynamique PHP
3. Optimiser les performances (cache)
4. Sécuriser avec HTTPS
5. Tester la montée en charge

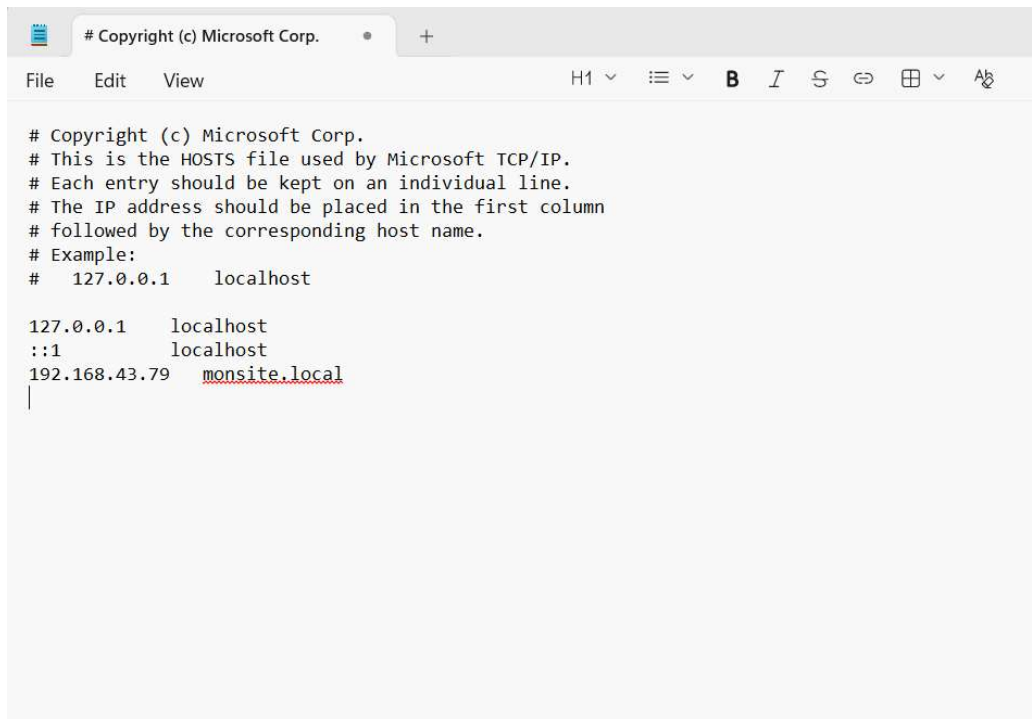
1- Modification du fichier hosts Windows

Sur Windows



```
C:\> Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Windows\System32\drivers\etc
C:\Windows\System32\drivers\etc>notepad/hosts
C:\Windows\System32\drivers\etc>_
```

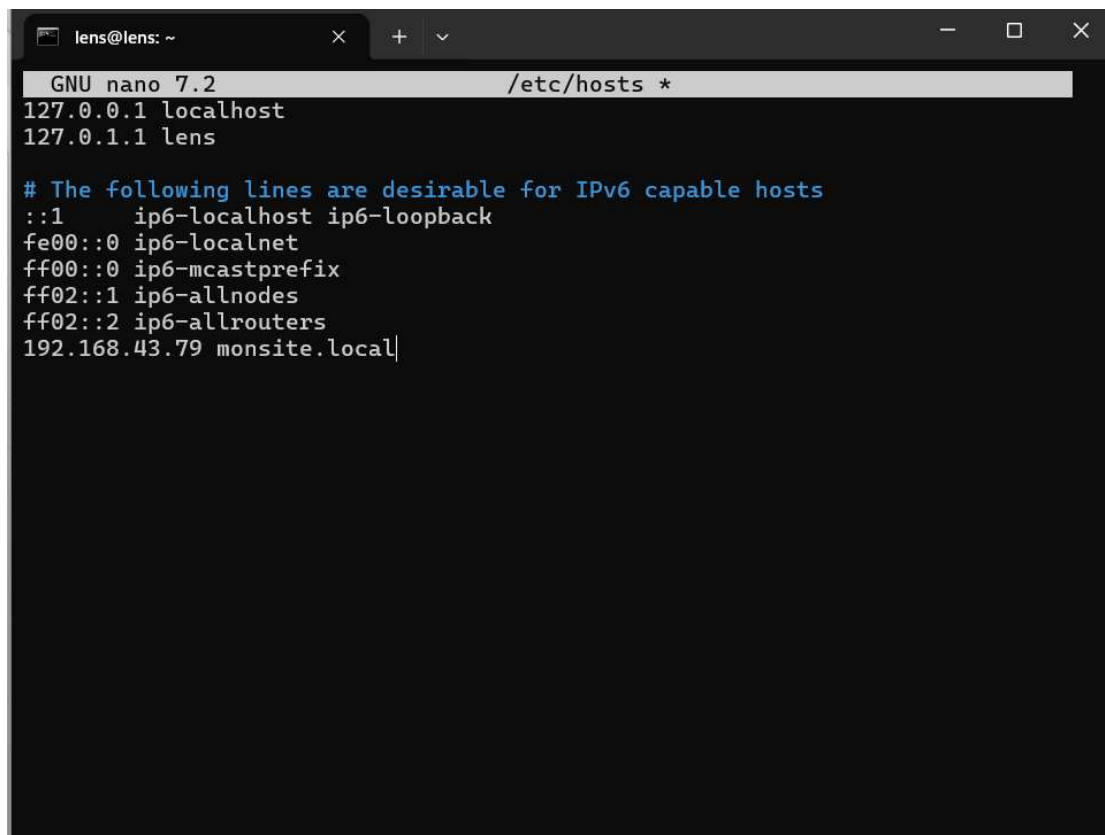


A screenshot of a Notepad window titled "# Copyright (c) Microsoft Corp.". The window contains the following text:

```
# Copyright (c) Microsoft Corp.
# This is the HOSTS file used by Microsoft TCP/IP.
# Each entry should be kept on an individual line.
# The IP address should be placed in the first column
# followed by the corresponding host name.
# Example:
# 127.0.0.1    localhost

127.0.0.1    localhost
::1         localhost
192.168.43.79  monsite.local
|
```

Sur le serveur



A screenshot of a terminal window showing the nano text editor editing the /etc/hosts file. The terminal output is as follows:

```
lens@lens: ~
GNU nano 7.2 /etc/hosts *
127.0.0.1 localhost
127.0.1.1 lens

# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
192.168.43.79 monsite.local|
```

2- Configuration du serveur WEB

-Installation des services et verification

```
lens@lens:~$ sudo apt install nginx php-fpm -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.6).
php-fpm is already the newest version (2:8.3+93ubuntu2).
0 upgraded, 0 newly installed, 0 to remove and 44 not upgraded.
lens@lens:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset>
   Active: active (running) since Mon 2026-04-13 11:48:36 UTC; 4h 40min a>
     Docs: man:nginx(8)
    Main PID: 855 (nginx)
      Tasks: 2 (limit: 2263)
     Memory: 3.6M (peak: 3.9M)
        CPU: 453ms
    CGroup: /system.slice/nginx.service
            └─855 "nginx: master process /usr/sbin/nginx -g daemon on; mas>
              └─858 "nginx: worker process"

Apr 13 11:48:36 lens nginx[810]: 2026/04/13 11:48:36 [warn] 810#810: duplic>
Apr 13 11:48:36 lens nginx[810]: 2026/04/13 11:48:36 [warn] 810#810: duplic>
Apr 13 11:48:36 lens nginx[810]: 2026/04/13 11:48:36 [warn] 810#810: duplic>
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Apr 13 11:48:36 lens nginx[810]: 2026/04/13 11:48:36 [warn] 810#810: duplic>
Apr 13 11:48:36 lens nginx[810]: 2026/04/13 11:48:36 [warn] 810#810: confli>
Apr 13 11:48:36 lens systemd[1]: Started nginx.service - A high performance>
lines 1-22/22 (END)
[9]+  Stopped                  systemctl status nginx

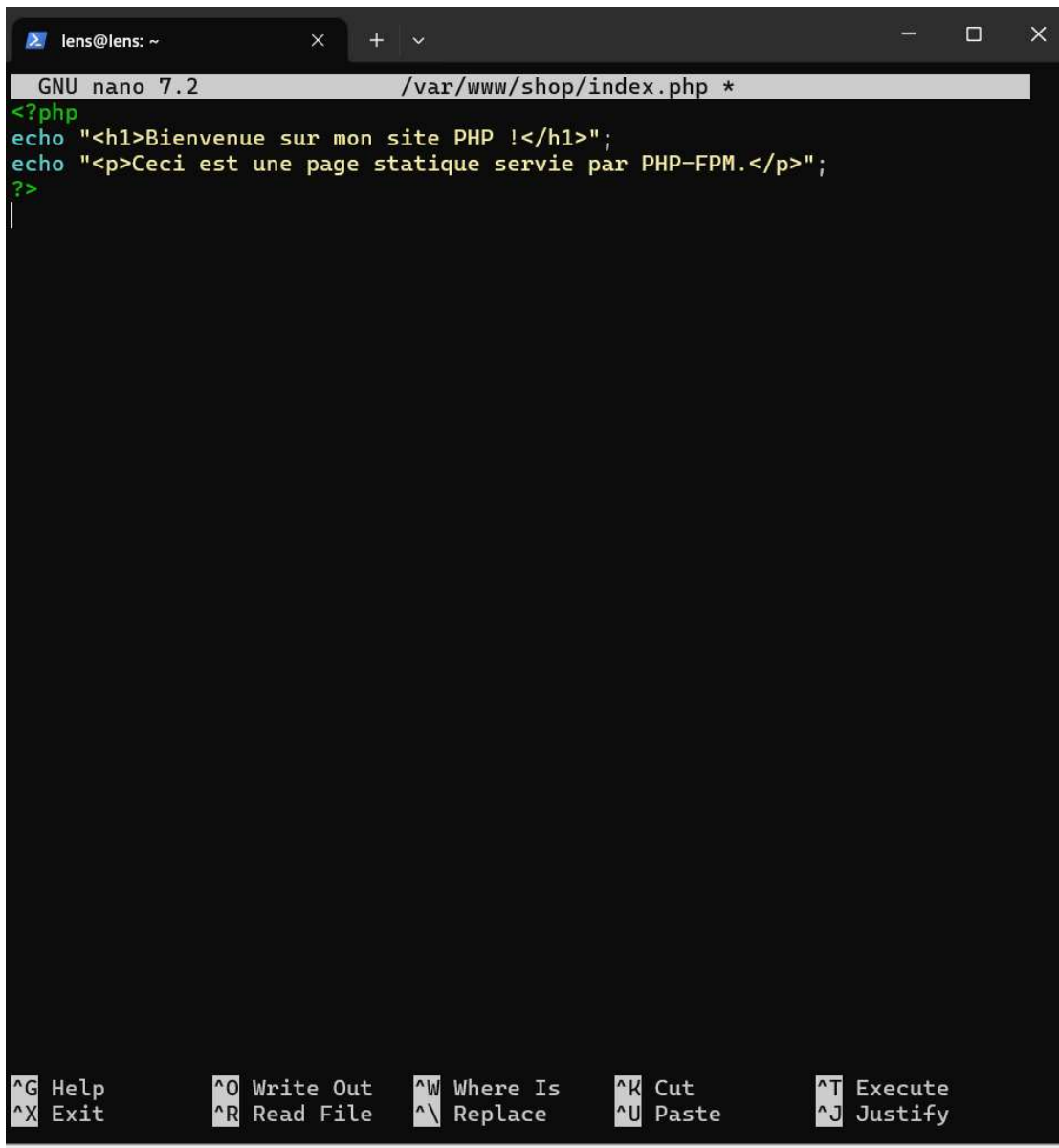
lens@lens:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service          enabled          enabled
phpsessionclean.service    static          -
phpsessionclean.timer      enabled         enabled
lens@lens:~$ sudo systemctl status php8.3-fpm
● php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager
```

```
[10]+ Stopped                                sudo systemctl status nginx

lens@lens:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service             enabled          enabled
phpsessionclean.service       static          -
phpsessionclean.timer         enabled         enabled
lens@lens:~$ sudo systemctl status php8.3-fpm
● php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php8.3-fpm.service; enabled; p>
   Active: active (running) since Mon 2026-04-13 16:11:18 UTC; 19min ago
     Docs: man:php-fpm8.3(8)
  Process: 19352 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install>
 Main PID: 19344 (php-fpm8.3)
    Status: "Processes active: 0, idle: 7, Requests: 3, slow: 0, Traffic: >
   Tasks: 8 (limit: 2263)
  Memory: 14.5M (peak: 14.9M)
     CPU: 487ms
    CGroup: /system.slice/php8.3-fpm.service
            └─19344 "php-fpm: master process (/etc/php/8.3/fpm/php-fpm.con>
              └─19345 "php-fpm: pool www"
                └─19346 "php-fpm: pool www"
                  └─19347 "php-fpm: pool zabbix"
                    └─19348 "php-fpm: pool zabbix"
                      └─19349 "php-fpm: pool zabbix"
                        └─19350 "php-fpm: pool zabbix"
                          └─19351 "php-fpm: pool zabbix"

Apr 13 16:11:18 lens systemd[1]: Starting php8.3-fpm.service - The PHP 8.3 >
Apr 13 16:11:18 lens systemd[1]: Started php8.3-fpm.service - The PHP 8.3 F>
lines 1-22/22 (END)
[10]+ Stopped                                sudo systemctl status php8.3-fpm
lens@lens:~$ |
```

3- Création du dossier du site et ajout du code



The image shows a terminal window with a dark background. At the top, a window title bar indicates the user is 'lens@lens: ~'. Below this, the terminal header shows 'GNU nano 7.2' and the file path '/var/www/shop/index.php *'. The main area of the terminal contains the following PHP code:

```
<?php
echo "<h1>Bienvenue sur mon site PHP !</h1>";
echo "<p>Ceci est une page statique servie par PHP-FPM.</p>";
?>
```

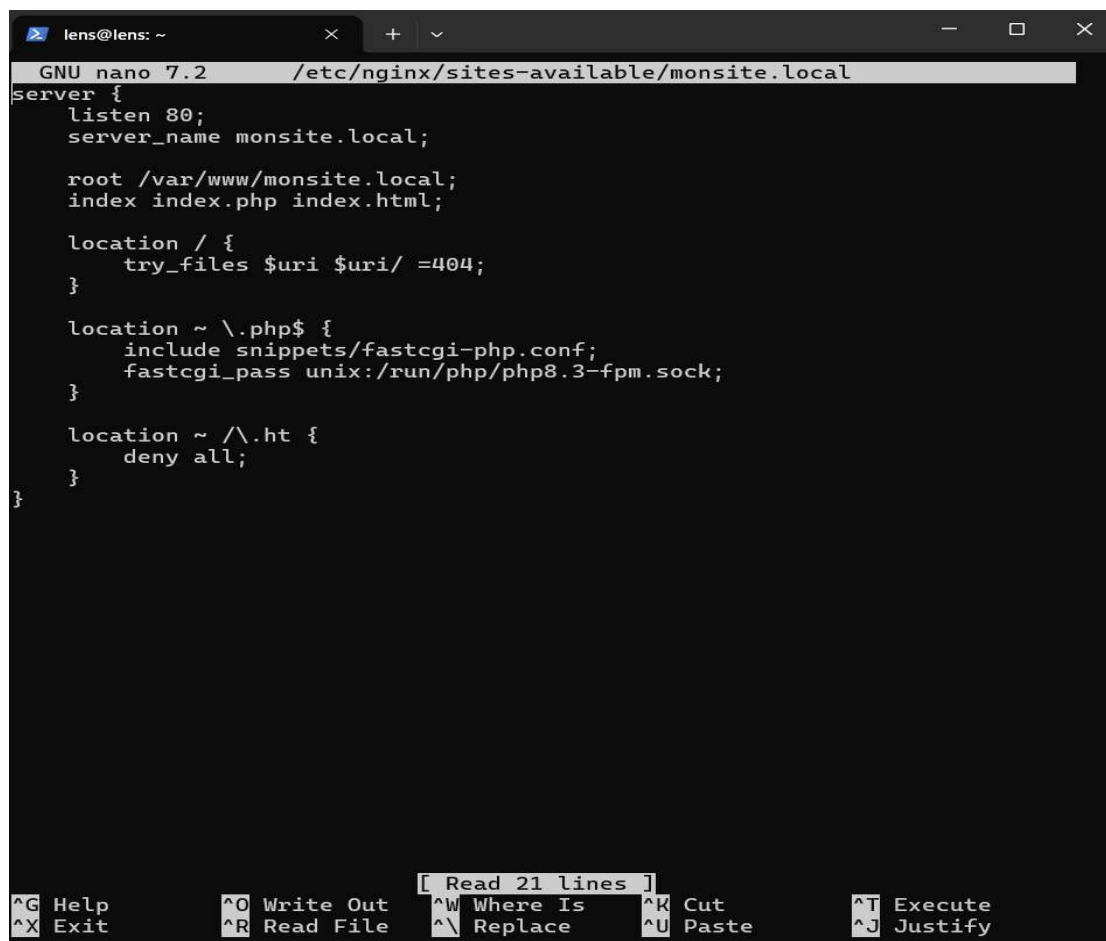
The code is written in a light green monospace font. A vertical cursor is positioned at the end of the third line. At the bottom of the terminal, a status bar displays various keyboard shortcuts for nano editor functions:

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify

4- Permissions

```
lens@lens:~$ sudo mkdir -p /var/www/shop
[sudo] password for lens:
lens@lens:~$ sudo nano /var/www/shop/index.php
lens@lens:~$ sudo chown -R www-data:www-data /var/www/
lens@lens:~$ sudo chmod -R 755 /var/www/
lens@lens:~$ |
```

5- Configuration de Nginx



The screenshot shows a terminal window with a nano editor editing the file `/etc/nginx/sites-available/monsite.local`. The configuration is for a server listening on port 80, serving files from `/var/www/monsite.local`. It includes a location for the root path and a location for PHP files using fastcgi_pass to connect to the PHP-FPM socket. The terminal window has a title bar with 'lens@lens: ~' and standard window controls. At the bottom, there is a status bar with various keyboard shortcuts like ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^X Exit, ^R Read File, ^_ Replace, ^U Paste, and ^J Justify. A message '[Read 21 lines]' is also visible in the status bar.

```
GNU nano '7.2' /etc/nginx/sites-available/monsite.local
server {
    listen 80;
    server_name monsite.local;

    root /var/www/monsite.local;
    index index.php index.html;

    location / {
        try_files $uri $uri/ =404;
    }

    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/run/php/php8.3-fpm.sock;
    }

    location ~ /\.ht {
        deny all;
    }
}
```

[Read 21 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^_ Replace ^U Paste ^J Justify

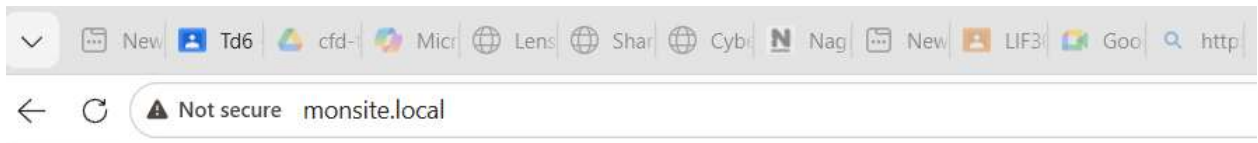
6- Activation du site

```
lens@lens:~$ sudo ln -s /etc/nginx/sites-available/shop /etc/nginx/sites-enabled/
ln: failed to create symbolic link '/etc/nginx/sites-enabled/shop': File exists
lens@lens:~$ sudo rm /etc/nginx/sites-enabled/default
rm: cannot remove '/etc/nginx/sites-enabled/default': No such file or directory
lens@lens:~$ sudo nginx -t
2026/04/13 21:50:40 [warn] 1510#1510: conflicting server name "monsite.local"
on 0.0.0.0:80, ignored
2026/04/13 21:50:40 [warn] 1510#1510: conflicting server name "192.168.43.79"
on 0.0.0.0:80, ignored
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
lens@lens:~$ sudo systemctl restart nginx
lens@lens:~$ |
```

7- Mise en place de Load Balancer

```
lens@lens: ~  
GNU nano 7.2 /etc/nginx/nginx.conf  
user www-data;  
worker_processes auto;  
pid /run/nginx.pid;  
  
events {  
    worker_connections 1024;  
}  
  
http {  
    include /etc/nginx/mime.types;  
    default_type application/octet-stream;  
  
    # Logs  
    access_log /var/log/nginx/access.log;  
    error_log /var/log/nginx/error.log;  
  
    # Sites activés  
    include /etc/nginx/sites-enabled/*;  
}  
  
[ Read 20 lines ]  
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute  
  
lens@lens:~$ sudo nginx -t  
2026/04/13 21:52:17 [warn] 1534#1534: conflicting server name "monsite.local"  
" on 0.0.0.0:80, ignored  
2026/04/13 21:52:17 [warn] 1534#1534: conflicting server name "192.168.43.79"  
" on 0.0.0.0:80, ignored  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
lens@lens:~$ sudo systemctl restart nginx  
lens@lens:~$ |
```

Test



Bienvenue sur mon site PHP

Avec HTTPS

Importer le nouveau certificat dans Windows

```
PS C:\Windows\system32> scp lens@192.168.43.79:/etc/ssl/myCA/monsite.local.crt "C:\Users\Lens Sandro Petiote\Desktop\monsite.local.crt"
lens@192.168.43.79's password:
monsite.local.crt                                     100% 1273   207.2KB/s   00:00
PS C:\Windows\system32> _
```

Test



Malgré l'importation du certificat je n'arrive pas a sécurisé mon site avec https

Cocclusion.

Ce TD m'a permis de savoir comment déployer une architecture web résiliente, configurer un site dynamique PHP, optimiser les performances (cache) et sécuriser avec HTTPS, donc la tâche a été réussie.